Battery Studio

OPERATION MANUAL FOR BMS HMI

Ver 1.0

Shanghai Richpower Electronics Co.Ltd.

2020

Contents

1. General Description....1
2. Software Installation....1
3. Quick Start....1
4. User Login....2
5. Load Protocol....2
6. System Configuration....3
7. Stand-alone Mode....4
8. Modify Parameter....5
9. Parallel Connection....7
10. View history data and realtime data....7
11. Calibration....8
12. Program upgrade....9
13. Remote Control Commands....10
14. Switch language....10

1. General Description

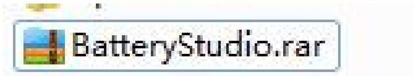
Battery Studio V4.3, which is developved by Shanghai Richpower Electronics Co.,Ltd., is an application software for monitoring and configuration for Lithium-ion battery management system(BMS). This software version integrates multiple functions and provides BMS users with one-stop service for monitoring and maintenence. It has key features as below.

- Supports all BMS modules from Richpower
- Supports live monitoring of all BMS data
- Supports Single & parallel mode
- Supports real-time/history data storage/read
- Support data analysis
- Support upload and download BMS parameters
- Support online BMS program upgrade
- Support calibration for voltage & current
- Support uultilingual interface

2. Software Installation

This software is a green version without installation. User can unzip the software package to required path, then click to run 'Battery Studio.exe'.

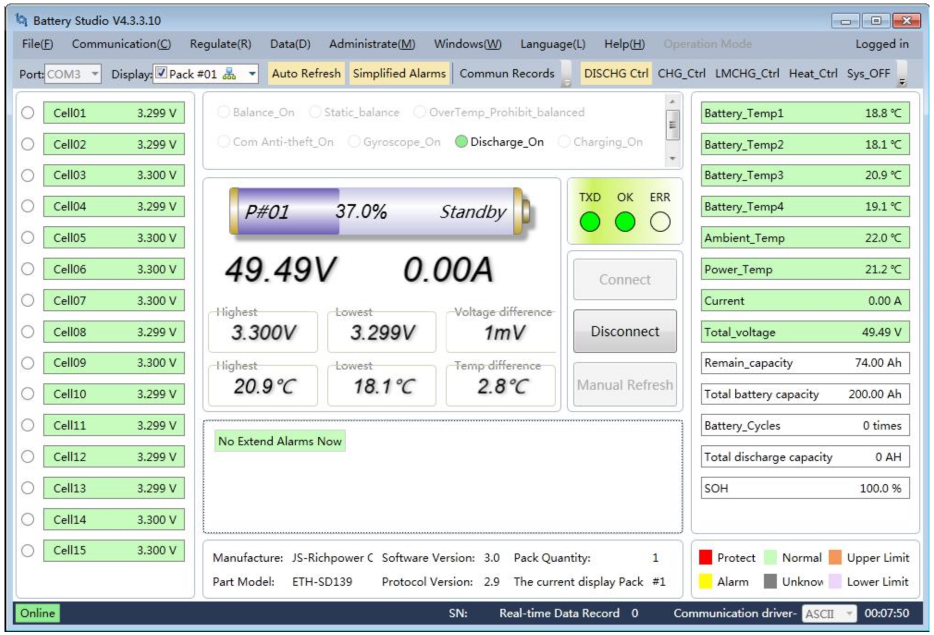
Unzip below file to computer.



3. Quick Start

Double click the exe file in the software path to run the software.

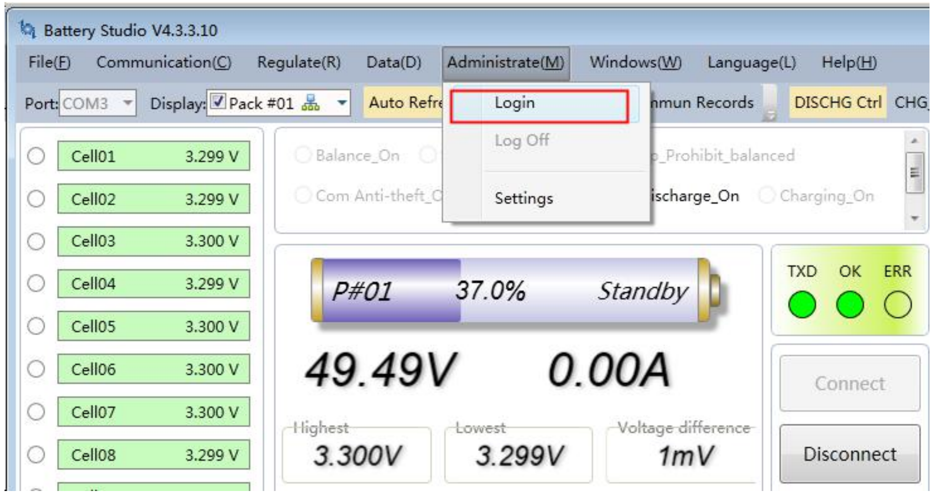
The software interface is as follows.



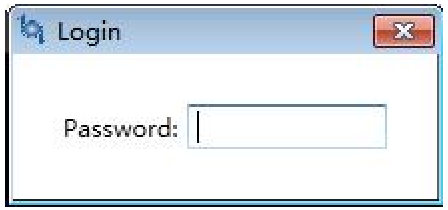
Connect BMS to computer via RS232 or RS485 serial port communication adapter, select correct port number, load corresponding BMS protocol provided by supplier, then click 'Connect' button on the HMI. The connection between BMS and HMI is then established.

4. User Login

Click 'Management'menu, select 'Login'option.

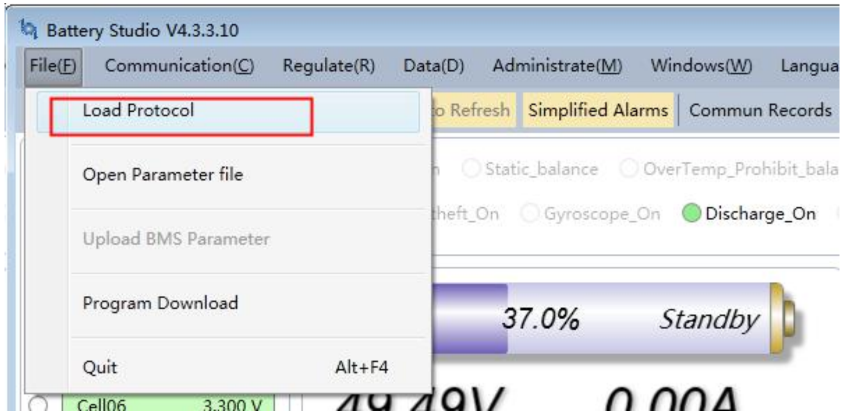


Input password 888888 in the popup input box and press Enter key to login. You are allowed to modify parameters and upgrade the BMS after login.



5. Load Protocol

Click 'File' menu, select 'Load protocol'option.



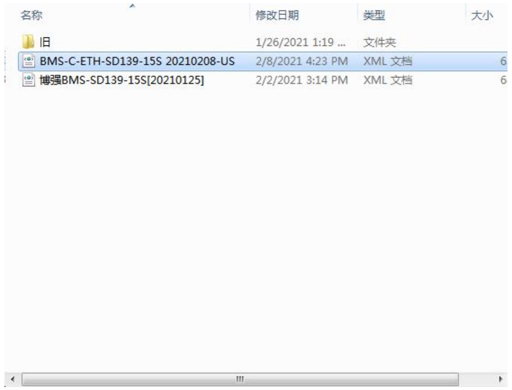
Select an option in the popup window.

Option 1: Automatically Load protocol online (realtime download from server).

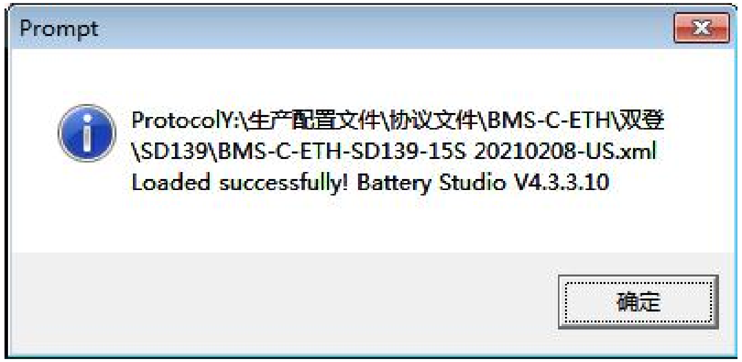
Option 2: Select BMS module number manually if you know it.

Option 3: Load local protocol if it's existed.

Click 'Confirm' button.

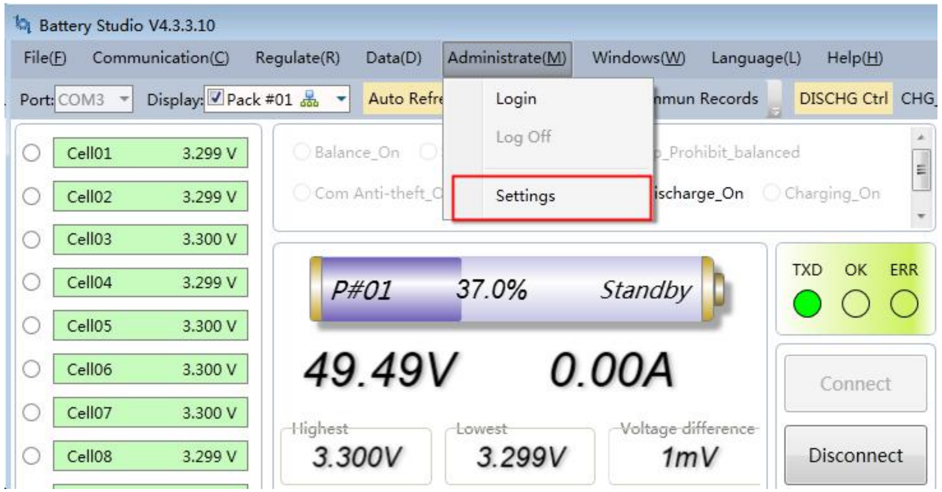


Protocol loaded sucessfully and you'll see below message. Click 'OK' to shutit.



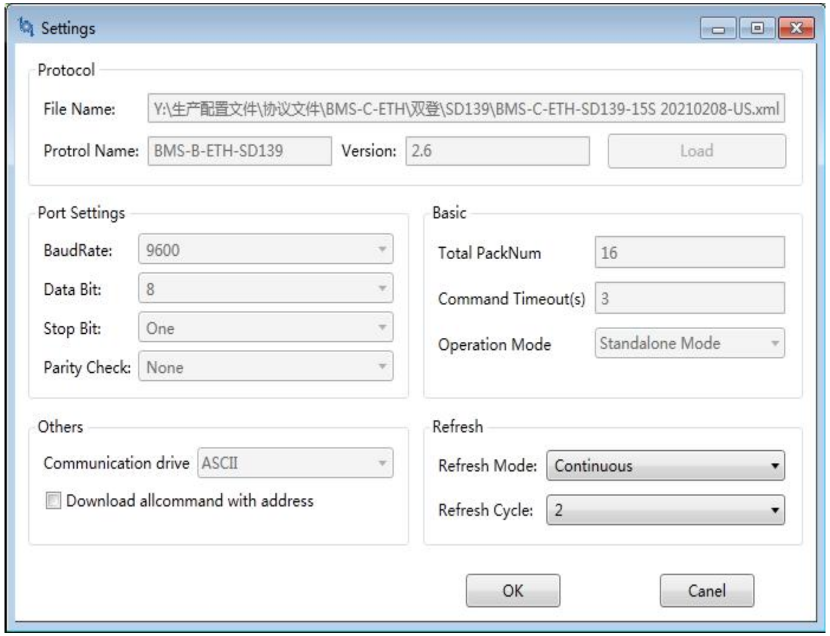
6. System Configuration

Click 'Management' menu and select 'Configuration' option.



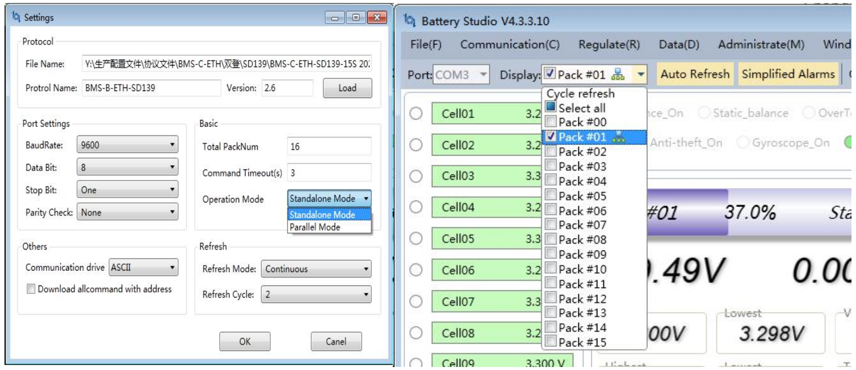
In the popup window you can find different configuration areas for setting the communication protocol, port, etc.

The communication protocol area has the same protocol loading function as described in Section 5. Serial communication parameters can be configured in the port setting area. The default configuration is 9600 baud rate, 8 data bits, 1 stop bit and no parity bit.

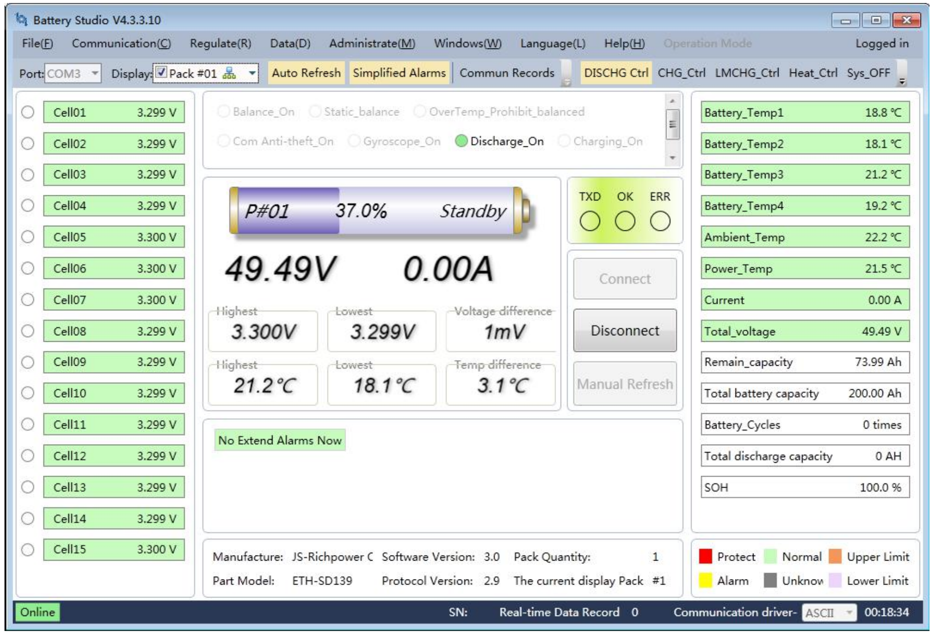


7. Stand-alone Mode

After loading correct protocol and system configuration, BMS can be connected to HMI. Click drop-down menu in software interface to select correct serial port, and click check box for 'Pack #01' in the drop-down menu next to it. Make sure the BMS is switched on and the dial address switch is set to address 0. Click 'Connect' button to connect BMS to HMI.



After the HMI is connected, the main interface of the software can display all kinds of data and status.

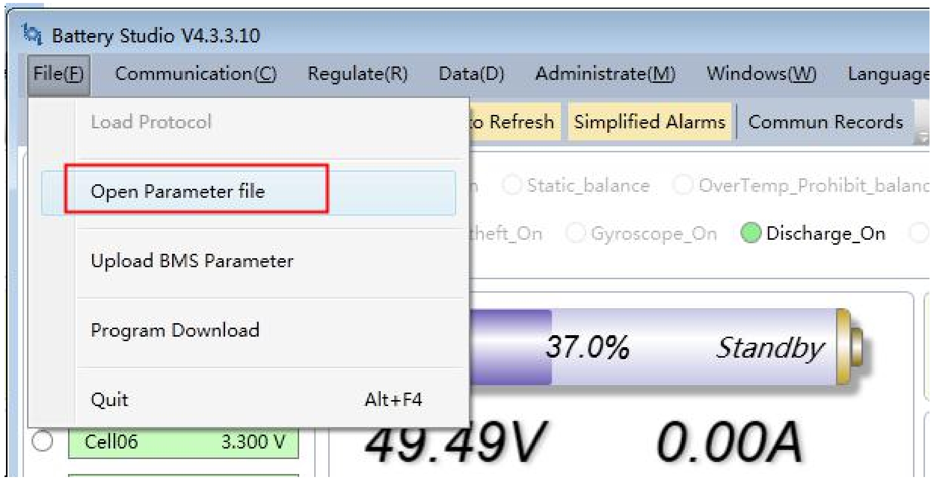


As shown in the figure above, the sampling voltage of each cell is displayed on the left side of the main interface, general voltage and current of the battery pack is displayed in the middle area, and the state of charge and battery pack status are displayed on the top. Temperature, voltage, current, capacity, cycle times and other information are displayed on the right side. At the bottom of the middle there is the alarm information box, in which all alarm protection information will be displayed.

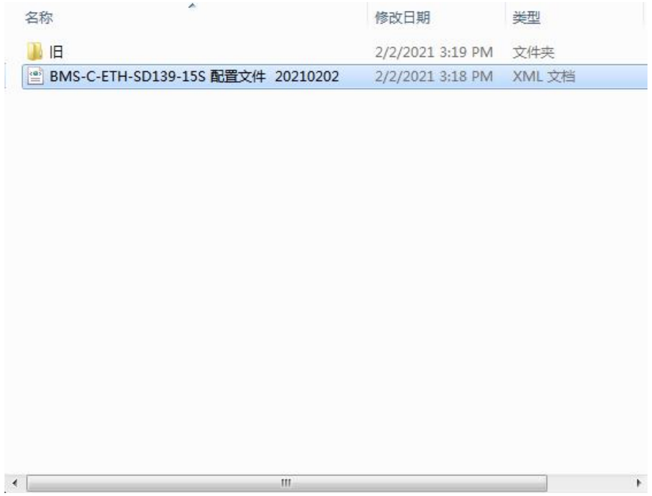
8. Modify Parameter

After connecting to the BMS and logging in the system, you can download the configured parameter file to BMS, or upload the BMS parameters for individual modification and then download them to BMS again. The method is as follows.

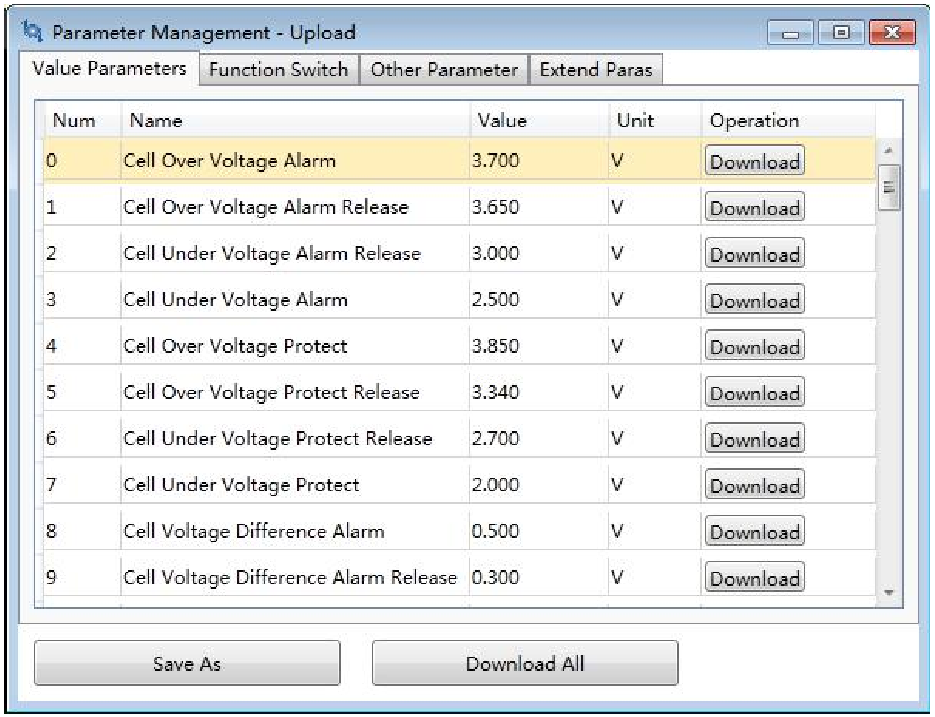
Click the file menu and select the "open parameter file" option.



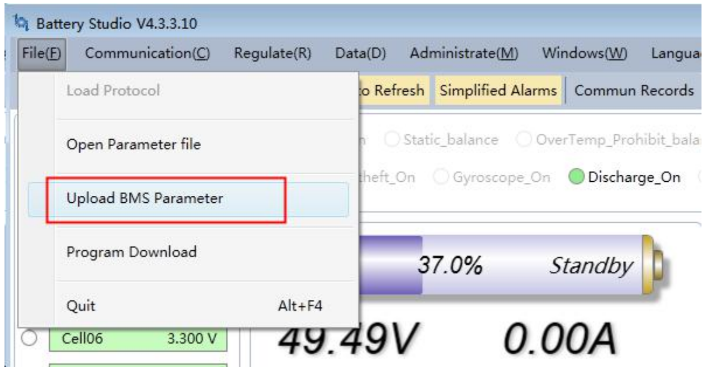
Select the correct configuration parameter file in the pop-up file selection box. Click the open button.



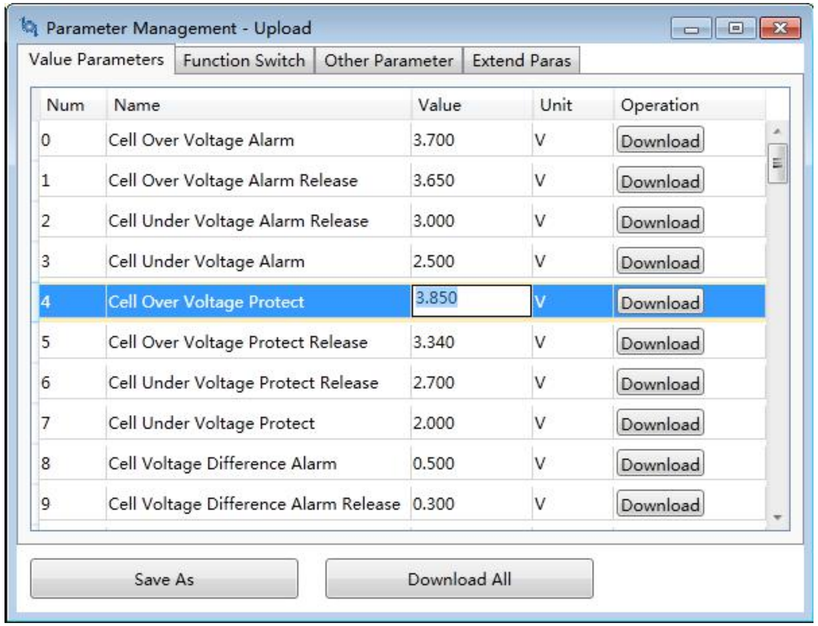
Check the settings and click "Download all parameters" button. The configuration parameters will be downloaded to BMS and become effective immediately.



Click the file menu and select 'upload BMS parameters' option.

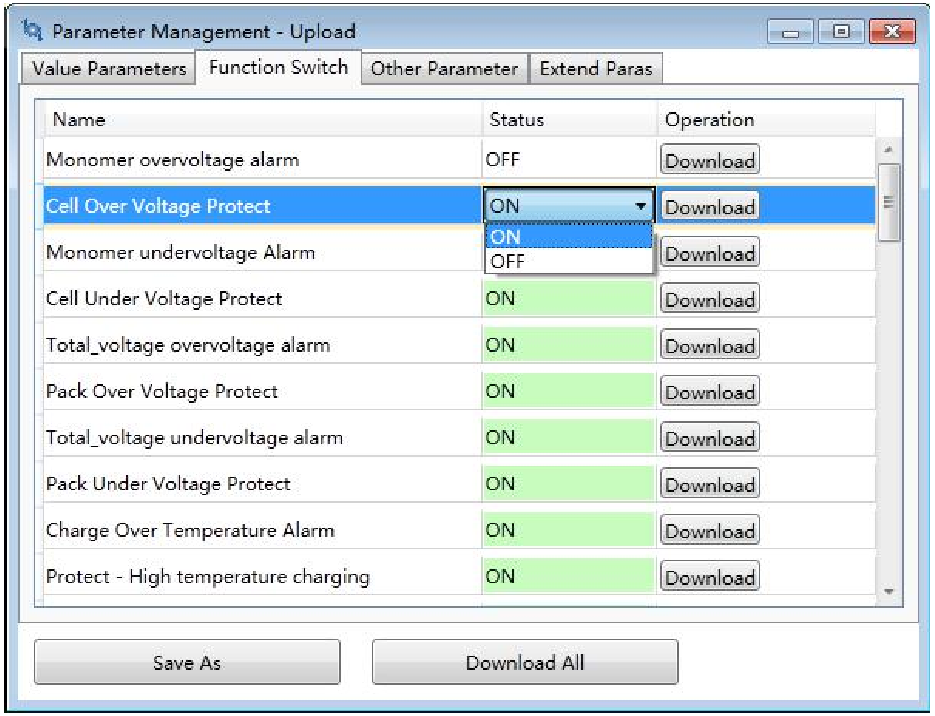


Thesoftware will read the configuration parameters in BMS and display them in the pop-up parameter management window.



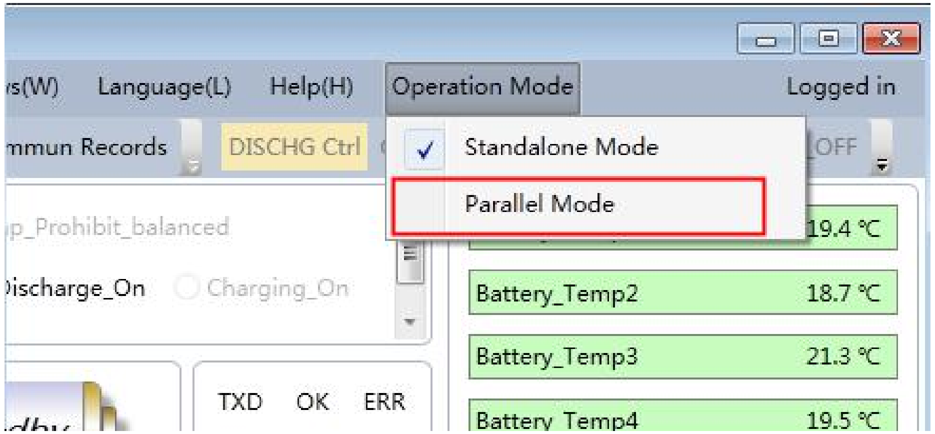
In the parameter management window, click the value you want to modify, enter appropriate parameters in the text box, and then click the "Download all" button below to download all parameters into BMS. If you click the "download" button on the right side of each parameter, only a single parameter will be downloaded into BMS, and the parameters modified in a single time cannot be saved after the BMS is powered off.

In addition to numerical parameters, you can also click the "function switch" tab in the window to operate and download the function switch. The method is the same as above.



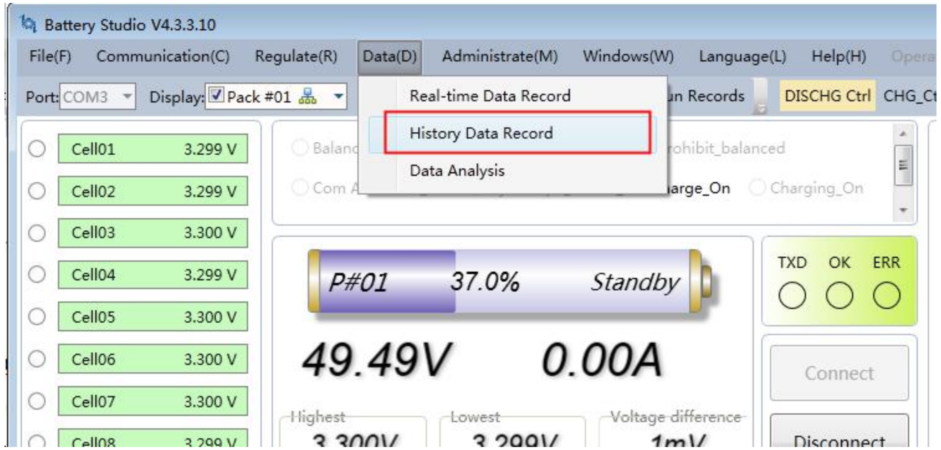
9. Parallel Connection

After login, the software will show 'Operation mode'in the menu. Select 'Parallel mode'and following the instructions. BMS can be connected via RS485 bus parallel connection. Set the dial address switch from address '2'. The total number of packs and specific pack numbers can be selected in the drop-down menu.

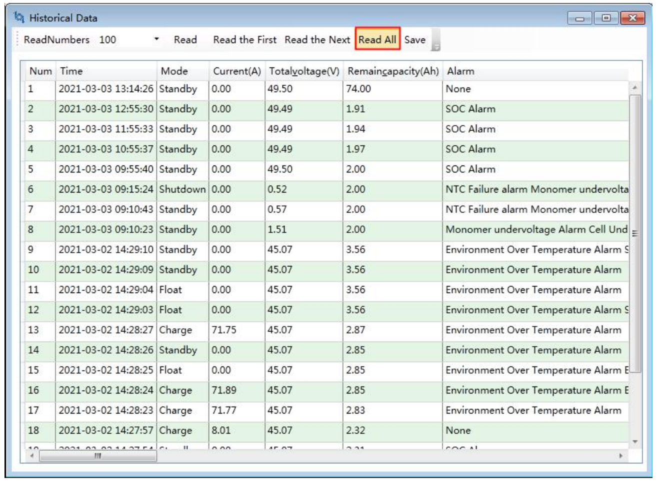


10. View history data and realtime data

Click the data menu and select the history data record option.

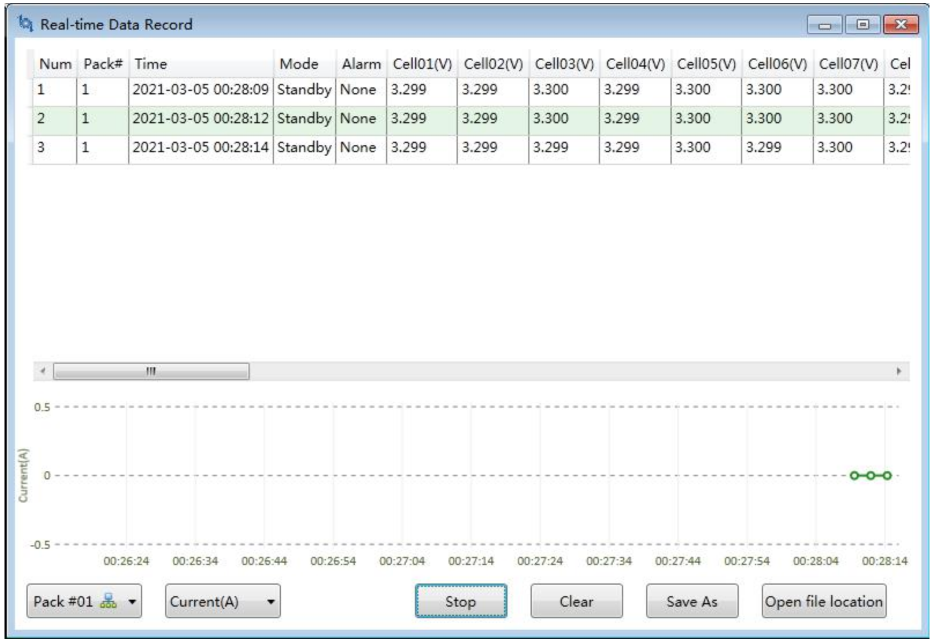


Click "read all records" button in the pop-up window, and the HMI will read all the history data stored in BMS and display them in the following window. Click Save button to export and save the record data to an excel file.



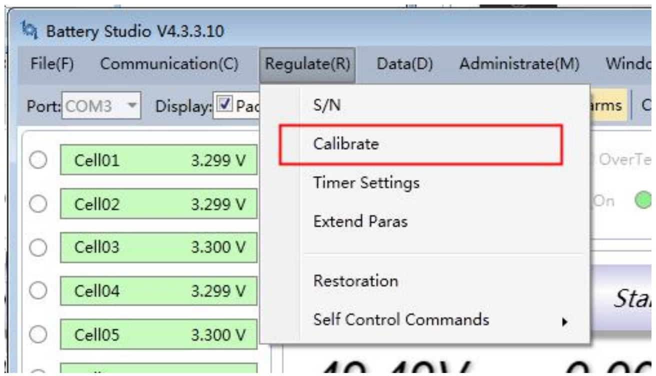
Num	Time	Mode	Current(A)	Totalvoltage(V)	Remaincapacity(Ah)	Alarm
1	2021-03-03 13:14:26	Standby	0.00	49.50	74.00	None
2	2021-03-03 12:55:30	Standby	0.00	49.49	1.91	SOC Alarm
3	2021-03-03 11:55:33	Standby	0.00	49.49	1.94	SOC Alarm
4	2021-03-03 10:55:37	Standby	0.00	49.49	1.97	SOC Alarm
5	2021-03-03 09:55:40	Standby	0.00	49.50	2.00	SOC Alarm
6	2021-03-03 09:15:24	Shutdown	0.00	0.52	2.00	NTC Failure alarm Monomer undervolta
7	2021-03-03 09:10:43	Standby	0.00	0.57	2.00	NTC Failure alarm Monomer undervolta
8	2021-03-03 09:10:23	Standby	0.00	1.51	2.00	Monomer undervoltage Alarm Cell Und
9	2021-03-02 14:29:10	Standby	0.00	45.07	3.56	Environment Over Temperature Alarm S
10	2021-03-02 14:29:09	Standby	0.00	45.07	3.56	Environment Over Temperature Alarm
11	2021-03-02 14:29:04	Float	0.00	45.07	3.56	Environment Over Temperature Alarm
12	2021-03-02 14:29:03	Float	0.00	45.07	3.56	Environment Over Temperature Alarm S
13	2021-03-02 14:28:27	Charge	71.75	45.07	2.87	Environment Over Temperature Alarm
14	2021-03-02 14:28:26	Standby	0.00	45.07	2.85	Environment Over Temperature Alarm
15	2021-03-02 14:28:25	Float	0.00	45.07	2.85	Environment Over Temperature Alarm E
16	2021-03-02 14:28:24	Charge	71.89	45.07	2.85	Environment Over Temperature Alarm E
17	2021-03-02 14:28:23	Charge	71.77	45.07	2.83	Environment Over Temperature Alarm
18	2021-03-02 14:27:57	Charge	8.01	45.07	2.32	None

Click data menu and select the "real-time data recording" option. HMI will read the real-time data of BMS and scroll in the window. Click the pause button to export and save the record data as Excel file by clicking the Save button.

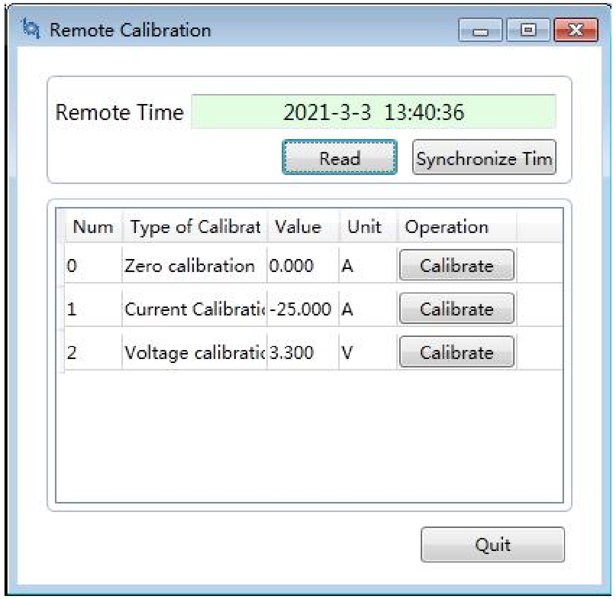


11. Calibration

Click ‘Control’ menu and select ‘Calibration’ option.



In the pop-up window, you can read the BMS time, synchronize the local time to BMS, calibrate the zero current of BMS, current calibration, voltage calibration, etc.

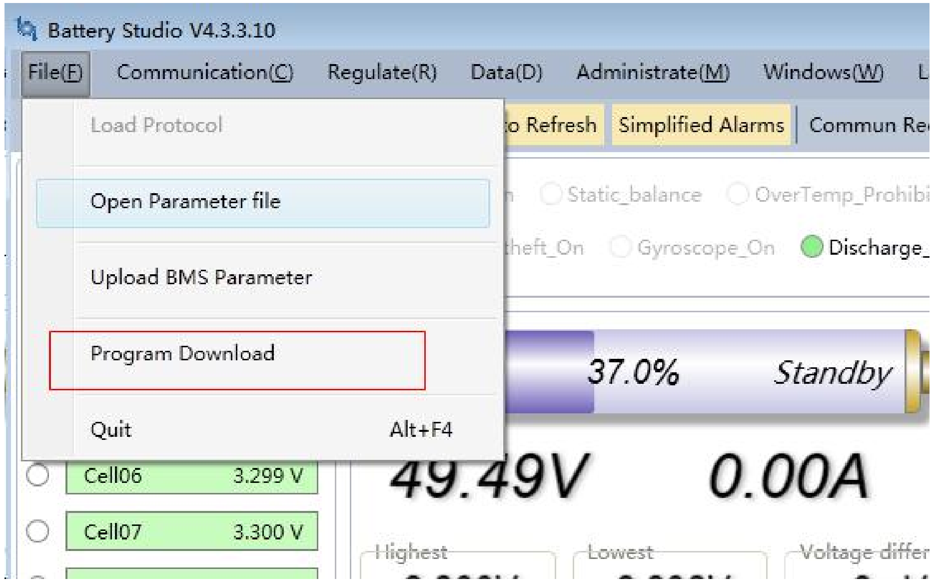


It should be noted that the zero point calibration should be carried out without charging and discharging current; the current calibration should be carried out under the 0.5c precise current discharge state; the voltage calibration should be carried out under the sampling voltage of 3.300v for each BMS cells.

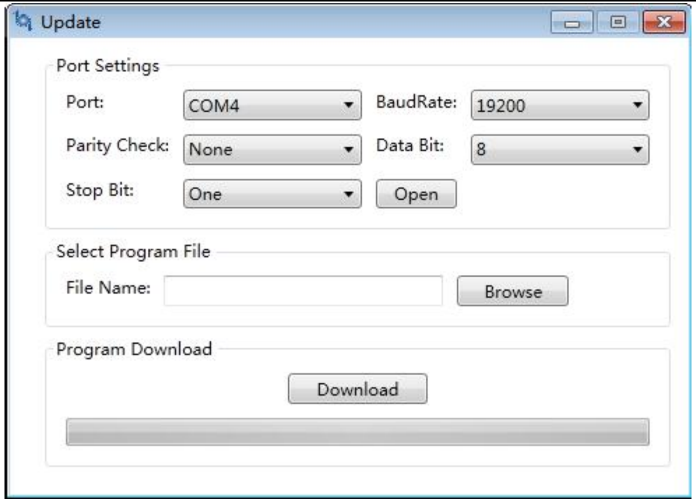
12. Program upgrade

The HMI supports downloading the upgrade program into BMS through 232 or 485 serial port. Usually, the BMS for shipment is a 485 boot program, which needs to be connected with 485 serial port.

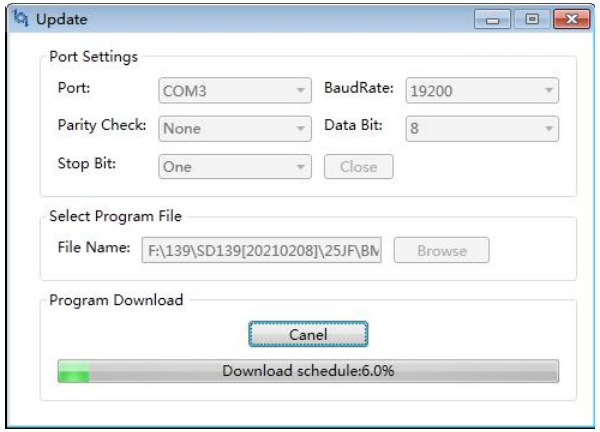
Click ‘file’ menu and select the program download option.



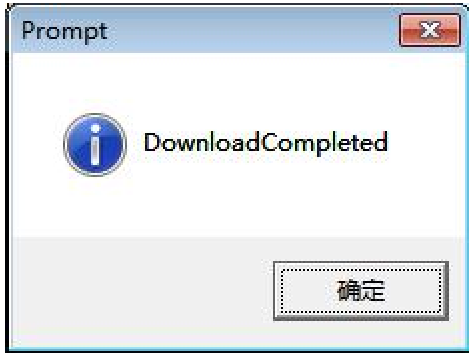
In the pop-up update window, set the correct port and baud rate (9600 or 19200 depending on the specific model), click the "open" button to open the serial port, and press the "Browse" button to select and load the program file in ehex format.



Press and hold the BMS reset key for more than 8 seconds until the indicator light shows the flow light status. At this time, click the "download" button. When the download progress bar appears, you can release the reset key.

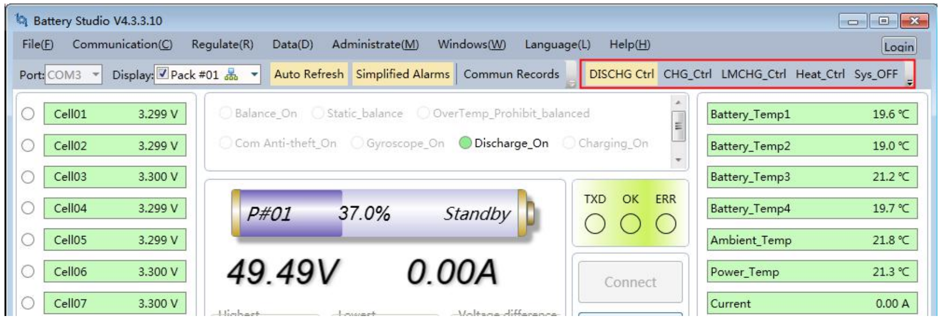


The following prompt will appear after the download is completed.



13. Remote Control Commands

In the upper part of the main interface there are several remote control command buttons, including discharge control, charging control, current limiting control, heating control, system shutdown, etc. Click the corresponding button to perform the corresponding action.



14. Switch language

Click language menu to select Chinese, English and Russian options. (the latest version is automatically adjusted according to the computer system language, no manual selection is required.)